

A

Multimodal data sources



Clinical labs (13 variables)

Age · Pack-years · dNLR · Albumin · ECOG ...
→ NLP sentence embedding (384-d)



Genomics (NGS, MSK-IMPACT)

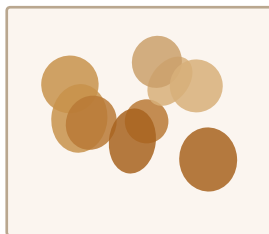
Oncogenic NSCLC driver mutations
and copy-number amplifications



L1 · L2 · L3

CT radiomics (PyRadiomics)

Per-lesion features by site:
primary (PC) · pleural (PL) · nodal (LN)

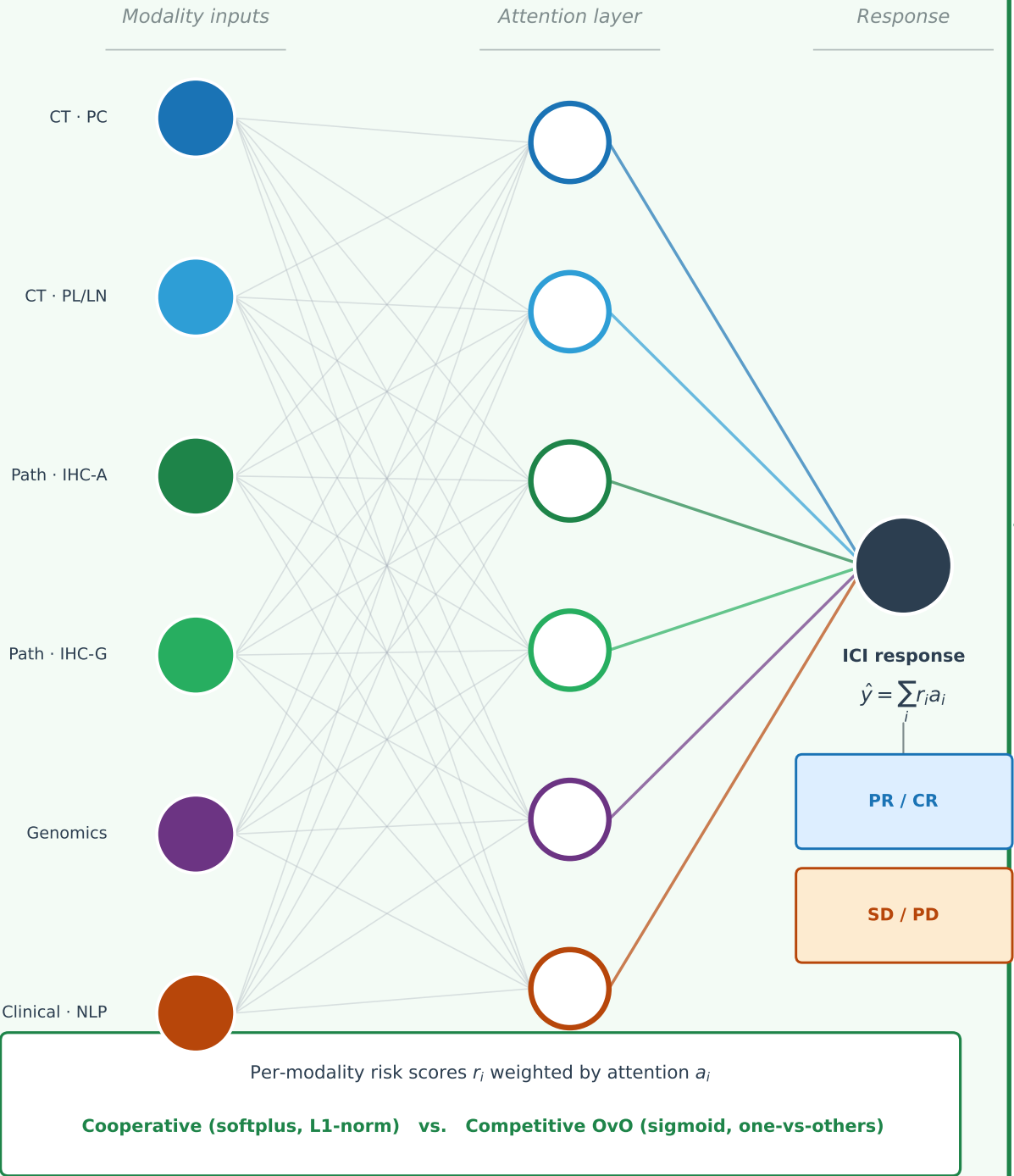


Pathology — PD-L1 IHC

PD-L1 texture (GLCM) features
PD-L1 % tumour positivity (TPS)

B

DyAM

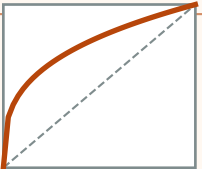


C

Evaluation

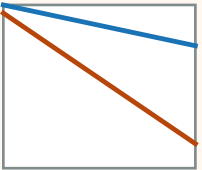
Binary classification

- ROC-AUC, DeLong 95% CI
PR/CR vs. SD/PD



Survival analysis

- Harrell's C-index
PFS discrimination
- Cox proportional hazards
forest plot, hazard ratios
- Kaplan-Meier
log-rank, risk strata



Modality combinations

Rad-only · Path-only
Rad + Genomics
Rad+Gen+Clinical (NLP)
Full (6 modalities)